K L Deemed to be University	CSE-1 -- KLVZA Course Handout 2025-2026, Even Sem

Course Title	**: DESIGN TOOLS WORKSHOP**
Course Code	**: 25ME1103**
L-T-P-S Structure	**: 0-0-4-0**
Pre-requisite	:
Credits	**: 2**
Course Coordinator	**: Atul Bhattad**
Team of Instructors	:
Teaching Associates	:
Syllabus :	CO-1: Office management and documentation: Technical tools: Typing and Keyboarding forBusiness Professionals: Accuracy and Speed-Building practice, Speed and Accuracy Tests, Demonstrateproficiency in sentence and paragraph creation. Office: Excel, Word, PowerPoint: Navigation & selectionshortcuts, Ribbon & Quick access tool bar, creating and using template, Text Formatting and saving file,Header & Footers, Working with bullets and numbered lists Understanding rows and columns, Naming Cell,Working with excel workbook and sheets, Creating slides and applying themes Latex Report Writing:Understanding Latex compilation, Basic syntax, writing equations, Tables, Figures handling, List of figures,List of tables, Generating index. Applications: Writing resume, Writing project reports Data Visualization:Introduction to data visualization, Data, types, Importance of data visualization, Different tools forvisualization, and comparisons in brief. Excel data explanation, Creation of column Chart, stacked bar chartand Heat map, Creation of excel dashboard. Creation of Dashboards in Power BI. Creation of bar charts,data tables, and pie charts in Power BI, creating slicers and maps in Power BI. CO-2: Basics of WebDevelopment: Introduction to a web browser and programming. HTML: Introduction to HTML, Creating asimple HTML page, HTML documents, Concept of tags, Basic structure of HTML document, Head, Body,Paragraph creation, line breaks, text, list, tables, Hyperlinks, and images. HTML5: Basic of HTML5,Special features of HTML5, image, audio, video. CSS: Creating Responsive web page with Bootstrap inCSS, color, background, margin, padding, text and font properties. Javascript: Introduction to Javascript,Variable, Keyword, Datatypes, Operator, Array, Conditional statement and Event handling. CO-3:Augmented Reality, Virtual Reality, Mixed Reality and Extended Reality: Introduction to AR, How ARworks, Difference between AR, VR and MR, Applications of AR and VR, AR components, AR TrackingSystem, Introduction to Markers, Markerless AR and Marker based AR, Introduction to VR, Head MountedDevices (HMD), Terrain Creation, 3D object creation. CO-4: Conceptual Design, Modelling andPrototyping: 3D modelling and printing: 2D Sketches to 3D Solid Model, Assembly modeling usingAUTODESK FUSION 360. Introduction to 3D printing, Part Model 3D printing through FDM process,Assembly Model 3D printing through FDM process. Electronic circuit building: Realization of bread boardand electronic components, Soldering in series and parallel connection, Verification of circuit laws: Ohmslaw, KCL and KVL law.

Text Books :	1. Rapid Prototyping: Principles and Applications" by Chua C.K., Leong, and Lim. C.S, 2ndEdition, World Scientific. 2. Augmented Reality and Virtual reality" by Timothy Jung, M.Claudia TomDieck, Springer. 3. Learn HTML & CSS by John Duckett. 4. HTML5 and CSS3 All-in-One for Dummies 5.Mastering Microsoft Power BI Expert techniques for effective data analytics and business intelligence byBrett Powell 6. LaTeX Tutorials: A Primer by Indian TeX Users Group by Indian TeX Users Group 7.Foulkes, Linda. Learn Microsoft Office 2019: A Comprehensive Guide to Getting Started with Word,PowerPoint, Excel, Access, and Outlook. United Kingdom, Packt Publishing, 2020.
Reference Books :	MS-Office. (2008). India: Laxmi Publications Pvt Limited. • Kottwitz, Stefan. LaTeXCookbook. Saint Lucia, Packt Publishing, 2015. • Duckett, Jon. HTML & CSS: Design and Build WebSites. Germany, Wiley, 2011. • Aspin, Adam. High Impact Data Visualization in Excel with Power View, 3DMaps, Get & Transform and Power BI. United States, Apress, 2016. • Cline, Lydia Sloan. Fusion 360 forMakers: Design Your Own Digital Models for 3D Printing and CNC Fabrication. United States, MakeCommunity, LLC, 2018. • Thurn, Laura, et al. 3D Printing: Understanding Additive Manufacturing.Germany, Carl Hanser Verlag GmbH & Company KG, 2018. • Augmented Reality and Virtual Reality: NewTrends in Immersive Technology. Germany, Springer International Publishing, 2021.
Web Links :	• https://www.coursera.org/learn/introduction-to-computers-and-office-productivity-software •https://www.latex-project.org/ • https://www.coursera.org/learn/html-css-javascript-for-web-developers •https://www.coursera.org/learn/data-analyze-visualize • https://www.coursera.org/learn/3d-model-creation-fusion-360 • https://www.coursera.org/specializations/3d-printing-additive-manufacturing •https://www.coursera.org/specializations/unity-xr? • https://www.w3schools.com/css/default.asp
MOOCS :	Web Links : • https://www.coursera.org/learn/introduction-to-computers-and-office-productivity-software https://www.coursera.org/learn/3d-model-creation-fusion-360 • https://www.coursera.org/specializations/3d-printing-additive-manufacturing •https://www.coursera.org/specializations/unity-xr? •
Course Rationale :	Course rationale to include the following aspects: To elaborate the reason why this course is included and towards what skills does this map To elaborate how the course will be offered mentioning if it is offered through a project based learning or through case-studies or through experiential learning.
Course Objectives :	To Demonstrate proficiency in typing sentence , paragraph , report , presentations along spread sheets using office tools, LaTeX tools and PowerBI. To Build a static website and blog with using html along with Special features of HTML5, CSS and Javascript. To Develop a virtual environment with cospace and construct a marker based Augumented Reality . To Utilise the softwares of Autodesk Fusion 360 and the same can be printed in 3D printer as physical prototype, Fundamentals of electrical circuit: Ohms law, KCL and KVL law
Global Certifications :	Fusion 360
Industry Specific Tools :	Fusion 360
Sector Skill Council :	3D printing, Virtual reality and Web development

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Demonstrate proficiency in typing sentence , paragraph , report ,presentations along spread sheets using office tools, LaTeX toolsand PowerBI	PO5,PSO2	2
CO2	Build a static website and blog with using html along withSpecial features of HTML5, CSS and Javascript	PO5,PO10	3
CO3	Develop a virtual environment with cospace and construct amarker based Augumented Reality	PO3,PSO1	3
CO4	Utilising the softwares of Autodesk Fusion 360 and the same canbe printed in 3D printer as physical prototype, Fundamentals ofelectrical circuit.	PO3,PSO1	3

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-1	COI-2	COI-3
CO1	2	**Btl-1** Memorize the office tools	**Btl-2** Show some advanced Officefunctions, including Word,Powerpoint and Excel andcompare same withPowerBI. Prepare LaTeXreport writing	
CO2	3	**Btl-1** Recall the web commands	**Btl-2** Demonstrate the web development tags	**Btl-3** Make use of Special features of HTML, CSS andJavascript web designing
CO3	3	**Btl-1** Remember the virtual tour	**Btl-2** Demonstrate the virtual scenario	**Btl-3** Develop an Virtual reality scenario
CO4	3	**Btl-1** Remember the modelling commands	**Btl-2** Understand the modelling and printing fundamentals	**Btl-3** Experiment with 3d Printer. Verification of electrical laws

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome
PO3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

Po No.	Program Outcome
PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PSO1	Ability to design, develop, and test software solutions using computational engineering principles.
PSO2	Ability to apply cutting-edge computing technologies to develop intelligent and efficient solutions.

Lecture Course DELIVERY Plan:
NO Delivery Plan Exists

Lecture Session wise Teaching – Learning Plan

No Session Plans Exists

Tutorial Course DELIVERY Plan:
NO Delivery Plan Exists

Tutorial Session wise Teaching – Learning Plan

No Session Plans Exists

Practical Course DELIVERY Plan:

Tutorial Session no	Topics	CO-Mapping
1	Introduction to Design tools workshop	CO1
2	Office Tools	CO1
3	Report wiritng using Latex	CO1
4	Preparing a resume using Latex	CO1
5	Excel dashboard	CO1
6	Dashboards in Power BI	CO1
7	Introduction to HTML	CO2
8	Line breaks, text, list, tables, Hyperlinks	CO2
9	Special features of HTML5, Canvas, audio, video	CO2
10	Creating Responsive web page	CO2

Tutorial Session no	Topics	CO-Mapping
11	Introduction to Javascript	CO2
12	Array	CO2
13	Sem In Exam -1	CO2
14	Introduction to VR	CO3
15	Applications of AR and VR	CO3
16	Creating Marker based	CO3
17	Creating Virtual environment in Cospace	CO3
18	Head Mounted Devices	CO3
19	Creating 3D tour	CO3
20	FUSION 360 INSTALLATION	CO4
21	3D modelling of any realistic model	CO4
22	3D PRINTING	CO4
23	Photogrammetry	CO4
24	Series and parallel connection	CO4
25	Verification of circuit laws	CO4
26	Lab In Sem-2 exam	CO4

Practical Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Introduction to Design tools workshop

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	2	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 2 Office Tools

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	2	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 3

Session Outcome: 1 Report wiritng using Latex

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
40	Execution	2	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 4

Session Outcome: 4 Preparing a resume using Latex

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	2	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 5

Session Outcome: 5 Excel dashboard

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	2	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 6

Session Outcome: 6 Dashboards in Power BI

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	2	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 7

Session Outcome: 7 Introduction to HTML

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	2	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 8

Session Outcome: 1 Line breaks, text, list, tables, Hyperlinks

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	2	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 9

Session Outcome: 1 Special features of HTML5, Canvas, audio, video

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	2	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 10

Session Outcome: 1 Creating Responsive web page

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	3	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 11

Session Outcome: 1 Introduction to Javascript

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	2	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 12

Session Outcome: 1 Array

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	3	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 13

Session Outcome: 1 Sem In Exam -1

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Sem In Exam -1	3	PPT	--- NOT APPLICABLE ---
40	Sem In Exam -1	3	PPT	--- NOT APPLICABLE ---
10	Viva	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 14

Session Outcome: 1 Introduction to VR

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	2	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 15

Session Outcome: 1 Applications of AR and VR

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
40	Execution	3	PPT	--- NOT APPLICABLE ---
10	Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 16

Session Outcome: 1 Creating Marker based

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	3	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 17

Session Outcome: 1 Creating Virtual environment in Cospace

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	3	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 18

Session Outcome: 1 Head Mounted Devices

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	3	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 19

Session Outcome: 1 Creating 3D tour

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	3	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 20

Session Outcome: 1 FUSION 360 INSTALLATION

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	2	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 21

Session Outcome: 1 3D modelling of any realistic model

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	3	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 22

Session Outcome: 1 3D PRINTING

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	3	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 23

Session Outcome: 1 Photogrammetry

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	3	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 24

Session Outcome: 1 Series and parallel connection

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	3	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 25

Session Outcome: 1 Verification of circuit laws

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Explanation	2	PPT	--- NOT APPLICABLE ---
40	Execution	3	PPT	--- NOT APPLICABLE ---
10	Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 26

Session Outcome: 1 Lab In Sem-2 exam

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
40	Lab In Sem-2 exam	3	PPT	--- NOT APPLICABLE ---
40	Lab In Sem-2 exam	3	PPT	--- NOT APPLICABLE ---
10	Viva	2	Talk	--- NOT APPLICABLE ---

Skilling Course DELIVERY Plan:
NO Delivery Plan Exists

Skilling Session wise Teaching – Learning Plan

No Session Plans Exists

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDEDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co

COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									
Mon	Theory	--	--	--	--	--	--	---	---	---
	Tutorial	--	--	--	--	--	--	---	---	---
	Lab	V-S29,V-S29,V-S29,V-S30,V-S30,V-S30,V-S31,V-S31,V-S31,V-S32,V-S32,V-S32	V-S29,V-S29,V-S29,V-S30,V-S30,V-S30,V-S31,V-S31,V-S31,V-S32,V-S32,V-S32	V-S9,V-S9,V-S9,V-S10,V-S10,V-S10,V-S11,V-S11,V-S11,V-S12,V-S12,V-S12	V-S9,V-S9,V-S9,V-S10,V-S10,V-S10,V-S11,V-S11,V-S11,V-S12,V-S12,V-S12	V-S21,V-S21,V-S21,V-S22,V-S22,V-S22,V-S23,V-S23,V-S23,V-S24,V-S24,V-S24	V-S21,V-S21,V-S21,V-S22,V-S22,V-S22,V-S23,V-S23,V-S23,V-S24,V-S24,V-S24	---	---	---
	Skilling	--	--	--	--	--	--	---	---	---
Tue	Theory	---	---	--	--	--	--	--	--	---
	Tutorial	---	---	--	--	--	--	--	--	---
	Lab	---	---	V-S9,V-S9,V-S9,V-S10,V-S10,V-S10,V-S11,V-S11,V-S11,V-S12,V-S12,V-S12	V-S9,V-S9,V-S9,V-S10,V-S10,V-S10,V-S11,V-S11,V-S11,V-S12,V-S12,V-S12	V-S21,V-S21,V-S21,V-S22,V-S22,V-S22,V-S23,V-S23,V-S23,V-S24,V-S24,V-S24	V-S21,V-S21,V-S21,V-S22,V-S22,V-S22,V-S23,V-S23,V-S23,V-S24,V-S24,V-S24	V-S29,V-S29,V-S29,V-S30,V-S30,V-S30,V-S31,V-S31,V-S31,V-S32,V-S32,V-S32	V-S29,V-S29,V-S29,V-S30,V-S30,V-S30,V-S31,V-S31,V-S31,V-S32,V-S32,V-S32	---
	Skilling	---	---	--	--	--	--	--	--	---
Wed	Theory	--	--	--	--	--	--	---	---	---
	Tutorial	--	--	--	--	--	--	---	---	---
	Lab	V-S25,V-S25,V-S25,V-S26,V-	V-S25,V-S25,V-S25,V-S26,V-	V-S1,V-S1,V-S1,V-S2,V-	V-S1,V-S1,V-S1,V-S2,V-	V-S13,V-S13,V-S13,V-S14,V-	V-S13,V-S13,V-S13,V-S14,V-	---	---	---

	Hour	1	2	3	4	5	6	7	8	9
		S26,V-S26,V-S27,V-S27,V-S27	S26,V-S26,V-S27,V-S27,V-S27	S2,V-S2,V-S3,V-S3,V-S3,V-S4,V-S4,V-S4	S2,V-S2,V-S3,V-S3,V-S3,V-S4,V-S4,V-S4	S14,V-S14,V-S15,V-S15,V-S15,V-S16,V-S16,V-S16	S14,V-S14,V-S15,V-S15,V-S15,V-S16,V-S16,V-S16			
	Skilling	--	--	--	--	--	--	---	---	---
Thu	Theory	--	--	--	--	--	--	---	---	---
	Tutorial	--	--	--	--	--	--	---	---	---
	Lab	V-S25,V-S25,V-S25,V-S26,V-S26,V-S26,V-S27,V-S27,V-S27	V-S25,V-S25,V-S25,V-S26,V-S26,V-S26,V-S27,V-S27,V-S27	V-S1,V-S1,V-S1,V-S2,V-S2,V-S2,V-S3,V-S3,V-S3,V-S4,V-S4,V-S4	V-S1,V-S1,V-S1,V-S2,V-S2,V-S2,V-S3,V-S3,V-S3,V-S4,V-S4,V-S4	V-S13,V-S13,V-S13,V-S14,V-S14,V-S14,V-S15,V-S15,V-S15,V-S16,V-S16,V-S16	V-S13,V-S13,V-S13,V-S14,V-S14,V-S14,V-S15,V-S15,V-S15,V-S16,V-S16,V-S16	---	---	---
	Skilling	--	--	--	--	--	--	---	---	---
Fri	Theory	--	--	--	--	--	--	---	---	---
	Tutorial	--	--	--	--	--	--	---	---	---
	Lab	V-S33,V-S33,V-S33,V-S34,V-S34,V-S34,V-S35,V-S35,V-S35,V-S36,V-S36,V-S36	V-S33,V-S33,V-S33,V-S34,V-S34,V-S34,V-S35,V-S35,V-S35,V-S36,V-S36,V-S36	V-S17,V-S17,V-S17,V-S18,V-S18,V-S18,V-S19,V-S19,V-S19,V-S20,V-S20,V-S20	V-S17,V-S17,V-S17,V-S18,V-S18,V-S18,V-S19,V-S19,V-S19,V-S20,V-S20,V-S20	V-S5,V-S5,V-S5,V-S6,V-S6,V-S6,V-S7,V-S7,V-S7,V-S8,V-S8,V-S8	V-S5,V-S5,V-S5,V-S6,V-S6,V-S6,V-S7,V-S7,V-S7,V-S8,V-S8,V-S8	---	---	---
	Skilling	--	--	--	--	--	--	---	---	---
Sat	Theory	--	--	--	--	--	--	---	---	---

	Hour	1	2	3	4	5	6	7	8	9
	Tutorial	--	--	--	--	--	--	---	---	---
	Lab	V-S33,V-S33,V-S33,V-S34,V-S34,V-S34,V-S35,V-S35,V-S35,V-S36,V-S36,V-S36	V-S33,V-S33,V-S33,V-S34,V-S34,V-S34,V-S35,V-S35,V-S35,V-S36,V-S36,V-S36	V-S17,V-S17,V-S17,V-S18,V-S18,V-S18,V-S19,V-S19,V-S19,V-S20,V-S20,V-S20	V-S17,V-S17,V-S17,V-S18,V-S18,V-S18,V-S19,V-S19,V-S19,V-S20,V-S20,V-S20	V-S5,V-S5,V-S5,V-S6,V-S6,V-S6,V-S7,V-S7,V-S7,V-S8,V-S8,V-S8	V-S5,V-S5,V-S5,V-S6,V-S6,V-S6,V-S7,V-S7,V-S7,V-S8,V-S8,V-S8	---	---	---
	Skilling	--	--	--	--	--	--	---	---	---
Sun	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--

REMEDIAL CLASSES:

Supplement course handout, which may perhaps include special lectures and discussions that would be planned, and schedule notified according

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCS

DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCS

EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Hours)	CO1	CO2	CO3	CO4
End Semester Summative Evaluation Total= 40 %	**Lab End Semester Exam**	Weightage	40		120	10	10	10	10
		Max Marks	100			25	25	25	25
In Semester Formative Evaluation Total= 25 %	**Continuous Evaluation - Project**	Weightage	12.5		120	3.125	3.125	3.125	3.125
		Max Marks	100			25	25	25	25
	Continuous Evaluation - Lab Exercise	Weightage	12.5		120	3.125	3.125	3.125	3.125
		Max Marks	100			25	25	25	25
In Semester Summative Evaluation Total= 35 %	**Lab In sem-II**	Weightage	17.5		120			8.75	8.75
		Max Marks	50					25	25
	Lab In sem-I	Weightage	17.5		120	8.75	8.75		
		Max Marks	50			25	25		

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course
In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Supplement course handout, which may perhaps include special lectures and discussions

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Supplement course handout, which may perhaps include special lectures and discussions

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Ramesh Kumar Sunkara	P	21-A	-	-	-	-
Suresh Kumar Grande	P	12-A	-	-	-	-
Narasimha Raju Kuthuri	P	17-A	-	-	-	-

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Nageswara Medikondu	P	2-A,25-A,29-A,33-A	-	-	-	-
Kali prasad	P	3-A,5-A,17-B,22-B	-	-	-	-
Venkata Durga Mareedu	P	2-C,12-C	-	-	-	-
Siddaiah Nalluri	P	19-C	-	-	-	-
Srinivasa Rao Seeram	P	7-A	-	-	-	-
Kanthimathi Tumuluri	P	11-A,14-A	-	-	-	-
Sai Sarath Kruthiventi	P	15-A,35-A	-	-	-	-
Buragadda Kumar	P	26-A,30-B	-	-	-	-
Banda Sandeep	P	16-B	-	-	-	-
Karimulla Syes	P	9-A	-	-	-	-
Raj Kumar Pittala	P	4-A	-	-	-	-
Sireesha Koneru	P	4-B,7-C,22-A	-	-	-	-
Agilesh R	P	15-B	-	-	-	-
MOHAN K.N	P	6-C	-	-	-	-
Atul Bhattad	P	1-A,6-A,10-A	-	-	-	-
Hasane Shaik	P	16-C	-	-	-	-
Sri Arigela	P	1-B	-	-	-	-
Priyaranjan Samal	P	16-A,21-B	-	-	-	-
Gopi Chatragadda	P	32-B	-	-	-	-
Gautam Hanumanthu	P	36-A	-	-	-	-
VENKATA SAI BOKKISAM	P	18-B,23-C	-	-	-	-
SIVANI PINNABOINA	P	10-B,24-B,27-B	-	-	-	-
NAGAMALLI ARASAVALLI	P	31-C	-	-	-	-
CHINTA MANJUSHA	P	19-B,24-C	-	-	-	-
Suneetha Emmela	P	5-C,9-C	-	-	-	-
Lakshmi Kuruguntla	P	20-B	-	-	-	-

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
SIVAPRASAD LEBAKA	P	26-C	-	-	-	-
TAMMINENI SREELATHA	P	3-B,26-B	-	-	-	-
APPIKATLA PHANI KUMAR	P	10-C	-	-	-	-
SRINIVASARAO ALLURI	P	11-C,29-B	-	-	-	-
RAJANIDEVI MERIGALA	P	8-B	-	-	-	-
RANJAN MAHAPATRA	P	8-C,29-C	-	-	-	-
PUTTAM RAO	P	18-A,27-A	-	-	-	-
Vineetha K V	P	6-B,11-B	-	-	-	-
GANESH BABU RAJENDRAN	P	22-C,27-C	-	-	-	-
Anand Dhanakotti	P	12-B	-	-	-	-
VIKRANT CHANDRAKAR	P	8-A,34-A	-	-	-	-
HARI CHANDANA PICHHIKA	P	15-C,33-C	-	-	-	-
CHANDAN KUMAR	P	1-C,20-A,25-B,31-B	-	-	-	-
Harsha Sri Kanakavalli	P	3-C	-	-	-	-
Raktim Guha	P	2-B,36-C	-	-	-	-
Sukanta Swain	P	5-B,23-A,32-C,33-B	-	-	-	-
Kaja Mohan	P	9-B,13-C,30-A,35-C	-	-	-	-
RAYABARAPU VENKATESWARLU	P	34-B	-	-	-	-
Pratap Dakua	P	7-B,14-C,18-C,31-A	-	-	-	-
Debabrata Bej	P	21-C	-	-	-	-
Diplesh Gautam	P	13-A,32-A,35-B	-	-	-	-

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Dr. K. Srinivasa rao	P	4-C,20-C,23-B,34-C	-	-	-	-
PARDHU THOTTEMPUDI	P	13-B,17-C,36-B	-	-	-	-
Mancharla Ravi	P	14-B,19-A,24-A,25-C,30-C	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes and carry the text book(s) or material(s) as prescribed by the Course Faculty to the class.

NOTICES

Most of the notices are available on the LMS platform.

All notices will be communicated through the institution email.

All notices concerning the course will be displayed on the respective Notice Boards.

Signature of COURSE COORDINATOR

(Atul Bhattad)(6072)(DBES-2)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of DBES-2

HEAD OF DEPARTMENT:

Approval from: DEAN-ACADEMICS
(Sign with Office Seal)
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